

Sleep Stage Classification from Wrist Accelerometer Alone

Quantifying the motion-only ceiling for four-class sleep staging, and what heart rate contributes

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github.com/ameyypawar/sleep-accel-dl

Abstract

Consumer fitness bands that lack a photoplethysmography sensor can measure motion but not heart rate, and every established wrist-wearable sleep staging model uses both together. This project isolates the accelerometer and asks what motion alone can achieve for four-class staging (Wake, Light, Deep, REM), then reintroduces heart rate to measure its marginal contribution. A 1D convolutional encoder with temporal attention feeds a bidirectional LSTM over 20-epoch context windows, trained on 31 subjects from the Walch et al. (2019) dataset with concurrent polysomnography. Under subject-wise five-fold cross-validation, accelerometer-only staging reaches Cohen's $\kappa = 0.101$, far below the 0.40 threshold for clinically fair agreement. Adding heart rate raises this to $\kappa = 0.346$, a paired-bootstrap improvement of +0.244 (95% CI +0.202 to +0.284). Critically, the reverse does not hold: adding the accelerometer to heart rate yields +0.018 (95% CI -0.005 to +0.042), an interval containing zero. The founding hypothesis is therefore rejected. However, a per-class analysis reveals that the two modalities are complementary along different axes: motion is the superior sleep/wake discriminator (binary κ 0.419 against 0.242 for heart rate), while heart rate supplies nearly all the information separating sleep stages. The practical conclusion is that a sensor-limited band can measure sleep duration reliably but cannot report sleep architecture.

1. Introduction and motivation

Polysomnography (PSG) is the clinical reference standard for sleep staging, but it requires an overnight laboratory stay with electroencephalography, electro-oculography and electromyography. Consumer wearables offer a cheaper, longitudinal alternative, and the research literature on wrist-based staging is correspondingly large. Almost without exception, that literature combines an accelerometer with a photoplethysmography (PPG) heart rate sensor.

This creates an unexamined dependency. A large share of fitness bands in circulation carry an accelerometer and no PPG sensor, either for cost or for battery reasons. If motion alone were sufficient, sleep architecture monitoring would extend to a substantially larger installed base at no hardware cost. The question this project answers is whether that is so.

The contribution is not the demonstration that motion carries some signal, which is well established for binary sleep/wake actigraphy. It is the isolation and quantification of the *motion-only ceiling* at four-class resolution, together with a controlled measurement of what heart rate adds on top.

1.1 Research questions

RQ1. Can a deep sequence model using only triaxial accelerometer input achieve Cohen's $\kappa > 0.40$, the conventional threshold for fair agreement, on four-class sleep staging?

RQ2. What is the marginal contribution of heart rate, and symmetrically, what does motion contribute once heart rate is available?

RQ3. Does modelling temporal context across consecutive epochs improve staging, or would a per-epoch classifier suffice?

These questions were formulated so that a negative answer to RQ1 remains an informative result. If motion proves insufficient, the ablation in RQ2 quantifies precisely what sensor-limited devices forgo.

2. Related work and positioning

Classical actigraphy algorithms such as Cole-Kripke and Sadeh operate on epoch-level activity counts and were designed exclusively for binary sleep/wake discrimination. They have no mechanism for distinguishing N3 from REM, both of which are characterised by near-absent gross motion.

Walch et al. (2019) collected the dataset used here and evaluated feature contributions including motion, heart-rate variability and a circadian "clock proxy", reporting approximately 72% accuracy on a *three*-class problem (Wake / NREM / REM) using shallow classifiers over hand-crafted features.

This work is positioned relative to that as follows, and makes no claim of priority. Three differences are material. First, the task here is *four*-class, separating Light from Deep sleep, which is the harder distinction and the one consumers are sold on. Second, the model operates on the *raw* triaxial waveform through a learned encoder rather than on engineered features. Third, and most importantly for the research question, no circadian clock proxy is used. A time-of-day prior is not a motion signal, and including one would confound any claim about what the accelerometer alone can do.

3. Data

The dataset is Walch et al. (2019), *Motion and heart rate from a wrist-worn wearable and labeled sleep from polysomnography*, distributed by PhysioNet under the Open Data Commons Attribution License v1.0. It comprises 31 subjects wearing an Apple Watch during concurrent clinical PSG, providing raw triaxial acceleration, PPG-derived heart rate, step counts, and expert-scored sleep stages at 30-second resolution.

The Multi-Ethnic Study of Atherosclerosis (MESA) cohort was considered first and rejected on technical grounds. MESA distributes actigraphy as 30-second *activity counts* rather than raw accelerometry, which makes the raw-waveform encoder central to this design impossible to train on it. It also requires institutional data-access approval, which is incompatible with a reproducible public repository.

3.1 Discrepancies between documentation and data

A format-probing script was written and executed *before* any parser, on the principle that undocumented structure should be measured rather than assumed. This proved necessary: the published description disagrees with the files in three respects, each of which would fail silently rather than raise an error.

Documented	Observed in the files
Stage codes {0,1,2,3,5}	Also 4 (356 epochs) and -1 (438 epochs)
(delimiter unstated)	motion/labels whitespace; heart_rate/steps comma
Approximately 50 Hz	10.0 to 66.6 Hz, varying by subject
(range unstated)	Timestamps reach -486,000 s (5.6 days pre-study)

The stage-4 discrepancy is the consequential one. Code 4 is Rechtschaffen and Kales stage 4, the deepest slow-wave sleep, which the AASM standard merged into N3 without renumbering. Following the documentation and treating it as unknown would have discarded 356 epochs, approximately 10% of the Deep class and the scarcest of the four. The mapping is implemented as a dictionary rather than a list indexed by stage code, since the code set is non-contiguous and positional indexing would misplace REM.

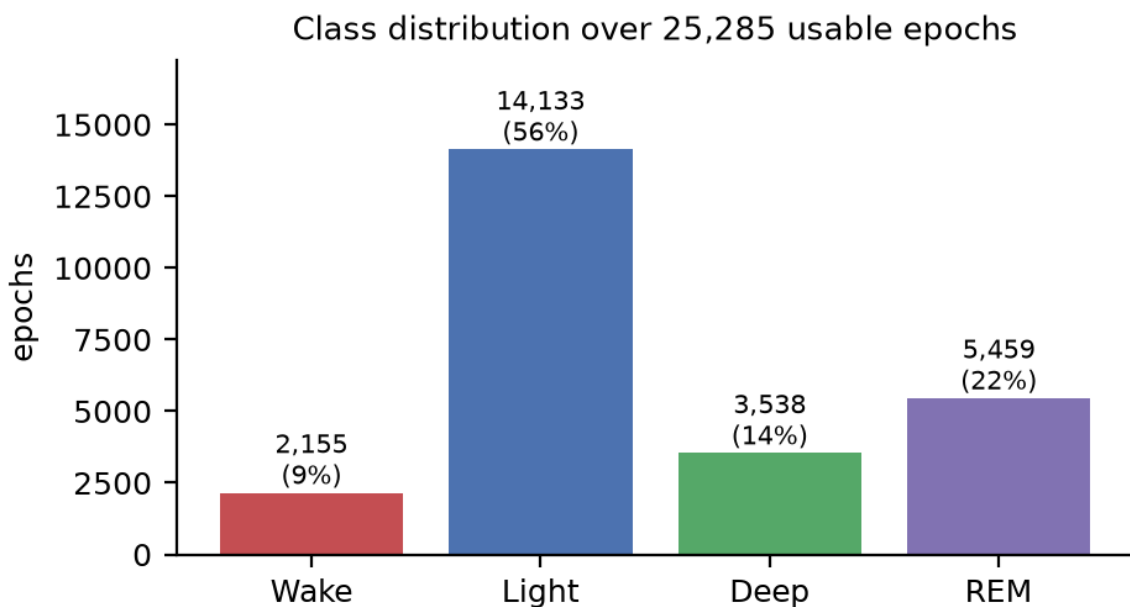


Figure 1. Class distribution after quality filtering. Light sleep dominates and Deep is scarcest, matching the expected architecture of a normal night. A uniform distribution here would have indicated label-signal misalignment.

3.2 Preprocessing and quality gating

The label file defines the epoch axis; accelerometer samples are resampled onto it. Deriving the axis from the accelerometer instead would allow a late-starting device to shift every epoch boundary relative to the PSG scoring, producing a constant misalignment no metric would detect.

Because the sampling interval is irregular and varies across subjects, resampling uses linear interpolation onto an explicit uniform 30 Hz grid. Functions such as `scipy.signal.resample` assume uniform input spacing and would return a time-distorted waveform without warning. A fourth channel, the vector magnitude, is appended because it is invariant to watch orientation on the wrist.

Each epoch is gated on two quality measures before interpolation: the fraction of the window spanned by real samples, and the largest gap between consecutive samples including the lead-in and lead-out intervals. This matters more than it appears. Interpolating across a sensor dropout yields a smooth, near-zero-variance segment, which is precisely the signature of deep sleep; a model trained on such data learns that absent signal means deep sleep and reports an inflated score. Subject 7749105 contains no accelerometer samples whatsoever in 86% of its epochs, and the gate correctly rejects them. Overall, 25,285 of 27,211 epochs (92.9%) survive.

Heart rate is summarised per epoch as five aggregate features (mean, standard deviation, minimum, maximum, and deviation from the subject's median) rather than resampled as a waveform, because PPG readings arrive orders of magnitude more sparsely than acceleration. Epochs with no reading receive zeroed features together with an explicit validity flag supplied to the model as an additional input; without that flag, "no measurement" and "0 bpm" would be indistinguishable, and 0 bpm is a physiologically extreme value rather than a neutral one.

4. Method

The architecture reflects a specific hypothesis about the problem. Deep and REM sleep are both characterised by near-absent gross motion, so they are close to indistinguishable within a single isolated 30-second epoch. What differentiates them is position within the roughly 90-minute ultradian cycle. The model is therefore built to classify an epoch *in context*, not in isolation.

Stage	Operation	Output
Input	30 s triaxial acceleration + magnitude at 30 Hz	4 x 900
Encoder	3 x [Conv1d(k=7), BatchNorm, GELU, MaxPool, Dropout]	128 x 112
Pooling	Additive temporal attention, learned query	128
Context	BiLSTM over 20 consecutive epoch embeddings	20 x 256
Head	Dropout, Linear	20 x 4

Attention pooling collapses the time axis to a single embedding per epoch while weighting the moments that carry information, rather than averaging uniformly across a window that is mostly quiescent. The bidirectional LSTM then reads the sequence of epoch embeddings. For the `accel_hr` and `hr_only` variants, the heart-rate features are projected and concatenated to the epoch embedding before the context stage; for `hr_only` the convolutional encoder is not instantiated at all.

Loss is cross-entropy with epochs failing the quality gate masked out via `ignore_index`, weighted by inverse class frequency raised to the power 0.5. The exponent was measured rather than assumed: full inverse frequency (power 1.0) inverted the prior, causing the model to predict Deep for 9,946 epochs against Light's 6,498 when the true proportions are 14% and 55%, and driving accuracy below the majority-class baseline. The square root is the compromise that learns the minority classes without overriding the prior.

5. Experimental protocol

Cross-validation is subject-wise, and this is the single most important methodological property of the study. One night yields approximately 1,000 epochs from one individual, and consecutive epochs from the same wrist are far more similar to each other than to any other subject's data. An epoch-wise split would permit a model to score highly by identifying the wearer rather than learning anything transferable about sleep, and the resulting inflation would be invisible in every diagnostic. A leakage assertion therefore executes at the start of every training run rather than only under test, and the test suite deliberately corrupts a fold to confirm the assertion fires.

Splits are three-way. Validation subjects for early stopping are drawn from the training portion, because selecting checkpoints on held-out test subjects would leak them into model selection as surely as training on them would. Fold membership is serialised to disk and reused verbatim by every variant, so the ablation arms are comparable by construction rather than by assumption.

Evaluation windows tile without overlap, so each test epoch is predicted exactly once. Training windows use a shorter stride, which provides free augmentation; applying the same overlap at evaluation would predict interior epochs several times and boundary epochs once, silently reweighting the test set.

Cohen's κ is the primary metric because accuracy is uninformative under this class distribution: predicting Light universally yields 55% accuracy and $\kappa = 0$. Every result carries a degeneracy flag that fires when any class is never predicted or one class exceeds 95% of predictions. Checkpoints are selected on validation macro F1 for the same reason.

6. Results

Variant	κ (pooled)	κ across folds	Macro F1	Accuracy	Binary κ
Accel only, no context	0.078	0.078 ± 0.010	0.281	0.568	0.361
Accel only	0.101	0.105 ± 0.018	0.336	0.517	0.419
HR only	0.327	0.328 ± 0.045	0.493	0.580	0.242
Accel + HR	0.346	0.347 ± 0.038	0.537	0.591	0.443

Table 1. Pooled over all held-out epochs, 31 subjects, subject-wise five-fold cross-validation, identical folds across all variants. No run was flagged degenerate.

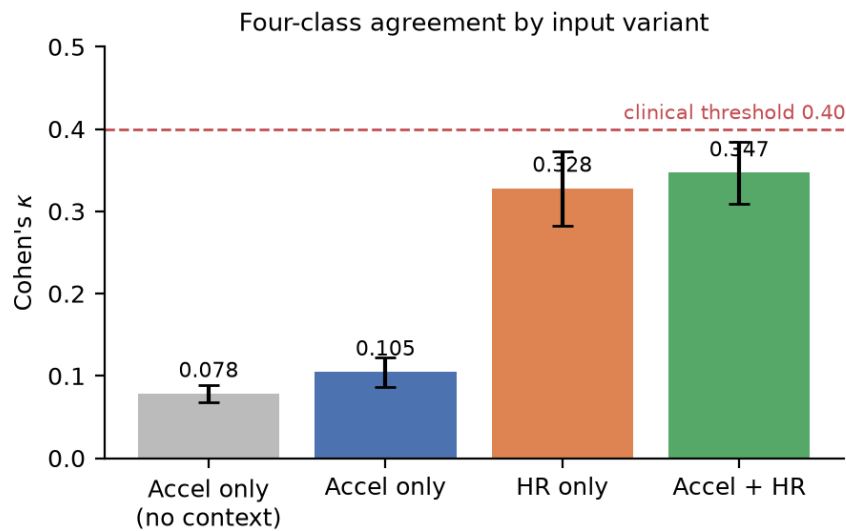


Figure 2. Four-class agreement by input variant, with standard deviation across folds. Only the heart-rate arms approach the clinical threshold, and none reaches it.

6.1 RQ1: motion alone is insufficient

Accelerometer-only staging reaches $\kappa = 0.101$ (0.105 ± 0.018 across folds), which is far below the 0.40 threshold for fair agreement. **RQ1 is answered in the negative.** The per-class breakdown localises the failure precisely: Deep reaches F1 0.090 and REM 0.105. Both stages are near-motionless, and without a cardiac signal the model has almost nothing by which to separate them.

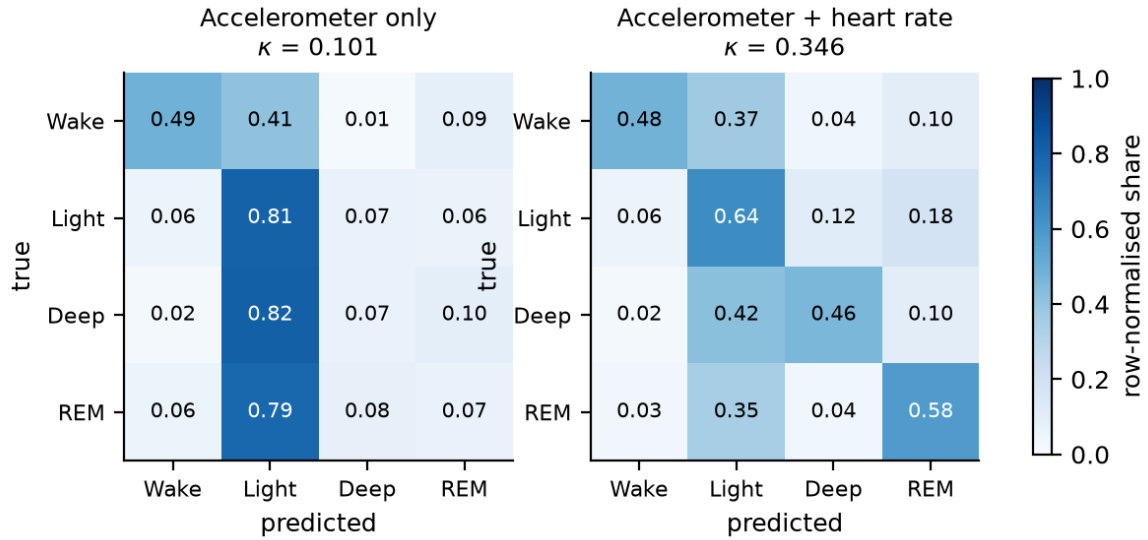


Figure 3. Row-normalised confusion matrices. Without heart rate, Deep and REM are largely absorbed into the adjacent classes; with it, both acquire substantial diagonal mass.

6.2 RQ2: heart rate carries the stage information

Adding heart rate to motion improves agreement by $+0.244$ (95% CI $+0.202$ to $+0.284$), an interval comfortably excluding zero. Heart rate alone already outperforms motion alone by $+0.226$ (95% CI $+0.180$ to $+0.271$).

The decisive comparison is the symmetric one. Adding the accelerometer to heart rate yields $+0.018$ (95% CI -0.005 to $+0.042$) — **an interval containing zero**. There is no statistically detectable benefit to supplying motion once heart rate is available, at least at this sample size. This is the strongest statement the dataset supports, and it contradicts the premise that a PPG-less device could perform four-class staging.

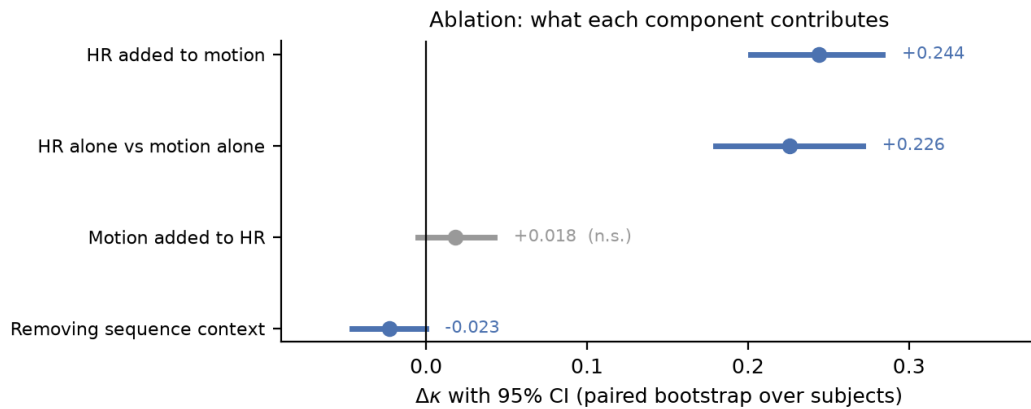


Figure 4. Paired bootstrap over subjects, 1,000 resamples. Subjects rather than epochs are resampled because epochs within one night are not independent, and the same resample is applied to both arms of each comparison; bootstrapping arms independently would widen the intervals and understate significance.

6.3 RQ3: temporal context contributes, narrowly

Removing the sequence model costs -0.023 (95% CI -0.046 to -0.000), an interval that only just excludes zero. The aggregate understates the effect, however. Without context the model predicts Deep and REM essentially never, with F1 of 0.000 and 0.001 respectively; it collapses onto the two classes that motion can detect. The BiLSTM is what enables any attempt at the stage distinctions at all, which is consistent with the hypothesis that motivated the architecture.

7. Discussion: the modalities are complementary along different axes

The aggregate κ conceals the most useful finding in this study. Motion and heart rate are not simply stronger and weaker versions of the same signal; they carry different information.

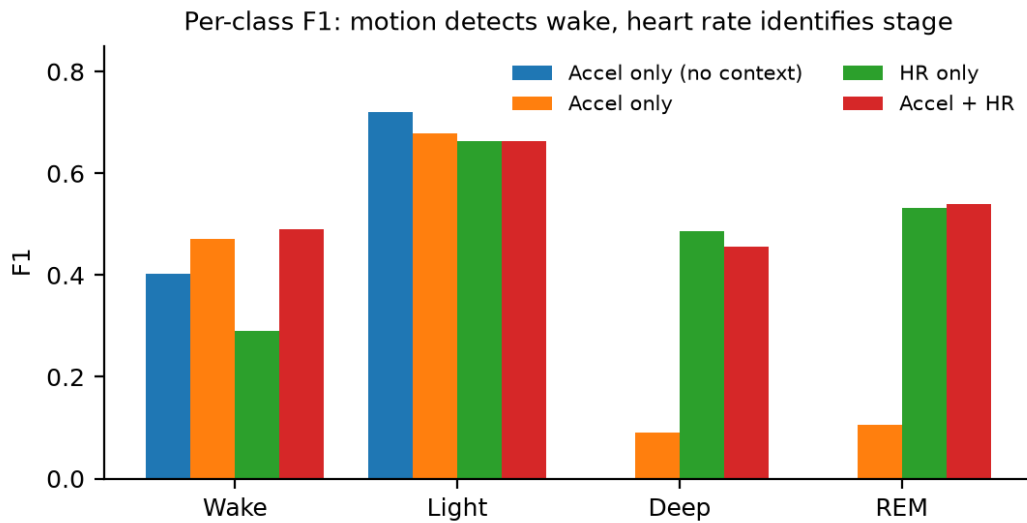


Figure 5. Per-class F1 across variants. Motion is the better wake detector; heart rate supplies the stage distinctions.

Variant	Wake F1	Light F1	Deep F1	REM F1	Binary κ
Accel only	0.471	0.679	0.090	0.105	0.419
HR only	0.291	0.664	0.487	0.531	0.242
Accel + HR	0.490	0.664	0.455	0.539	0.443

Table 2. The complementarity. Motion wins on Wake and on binary sleep/wake; heart rate wins decisively on Deep and REM.

Motion determines whether a person is awake. Accelerometer-only attains binary sleep/wake $\kappa = 0.419$ against 0.242 for heart rate alone. Motion is substantially the better wake detector, which is precisely why classical actigraphy has been clinically useful for decades.

Heart rate determines which stage of sleep. It raises Deep F1 from 0.090 to 0.487 and REM from 0.105 to 0.531, in a regime where motion is nearly blind.

Combining both yields the best binary κ (0.443) and the best macro F1 (0.537). The reason `accel_hr` barely exceeds `hr_only` on four-class κ is that the metric is dominated by the three sleep-stage distinctions, where motion contributes little — not because motion is uninformative in general. Reporting only the headline κ would have obscured this entirely.

Practical implication. A wearable with an accelerometer and no PPG sensor can report total sleep time and sleep efficiency with reasonable fidelity. It cannot report sleep architecture. Consumer devices that present deep-sleep and REM breakdowns without a cardiac sensor should be treated with scepticism.

8. Threats to validity

Sample size. 31 subjects is small. The confidence interval on the motion-added-to-heart-rate comparison is $[-0.005, +0.042]$, so a true effect of up to approximately +0.04 κ cannot be excluded. The correct reading is "no detectable benefit at this sample size", not "no benefit".

Single device and population. All data comes from one Apple Watch model in a single-site study. Sampling rates varying from 10 to 67 Hz across subjects suggest heterogeneous firmware or collection conditions even within this cohort.

Training budget. Each arm trained for eight epochs, selected on validation macro F1. Validation curves had plateaued by approximately epoch five in every fold, so the arms are not materially undertrained, but no hyperparameter search was performed and absolute performance would likely improve modestly with one.

Transductive heart-rate baseline. The per-subject heart-rate baseline is the median over the entire recording, including epochs that later fall in held-out folds. This is a deliberate choice, since the quantity is per-subject by construction and a full night is available before scoring on a real device, but it should be stated rather than discovered.

Class weighting interacts with the metric. The weighting exponent was found to move κ and accuracy in opposite directions. Results are reported at a single exponent (0.5) applied uniformly to every arm, so comparisons are internally consistent, but the absolute κ values are mildly sensitive to this choice.

9. Conclusion

Four-class sleep staging from wrist accelerometry alone reaches Cohen's $\kappa = 0.101$, well short of clinical usability, and adding motion to heart rate produces no statistically detectable improvement. The hypothesis that motivated this work — that sensor-limited fitness bands could deliver sleep architecture — is not supported by this dataset.

The more useful finding is the decomposition. Motion and heart rate are complementary along orthogonal axes: motion determines wakefulness, heart rate determines sleep stage. This explains both why classical actigraphy succeeds at its intended binary task and why extending it to four classes fails. It also delimits precisely what a PPG-less device can honestly claim.

The methodological contribution is a validation protocol built to make silent failure difficult: subject-wise splits with assertions that execute in production rather than only in tests, quality gating that rejects sensor dropouts before they can be interpolated into convincing artefacts, a degeneracy flag on every reported metric, and a format probe that measured the data rather than trusting its documentation — which proved necessary in three separate respects.

10. Reproducibility

All code, pinned dependencies and the analysis pipeline are public. The test suite comprises 71 tests, all of which execute without the dataset present, and specifically cover each silent-failure mode described above.

```
git clone https://github.com/ameyypawar/sleep-accel-dl
uv venv --python 3.11 .venv && source .venv/bin/activate
uv pip install -r requirements.txt
pytest -q
python scripts/probe_raw_format.py
python -m sleepaccel.data.build_cache
./scripts/run_all.sh && python scripts/ablation.py
streamlit run app/streamlit_app.py
```

An interactive dashboard presents the dataset summary, training curves, confusion matrices, the ablation comparison, and a per-subject hypnogram viewer aligning predictions against PSG ground truth.

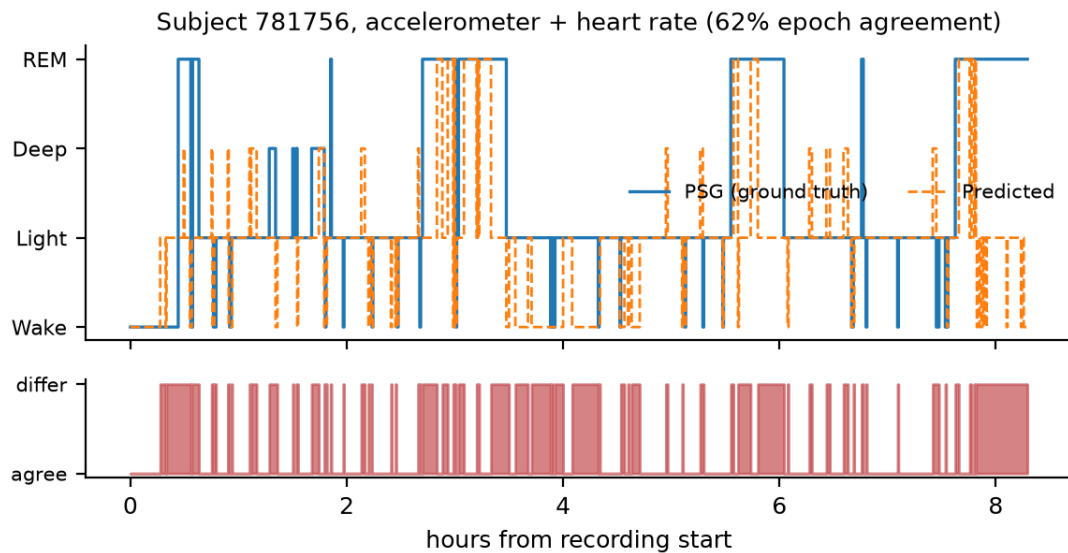


Figure 6. Predicted hypnogram against polysomnography for the longest recording, accelerometer plus heart rate. Disagreement concentrates at stage transitions, where inter-scorer agreement is also lowest.

References

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